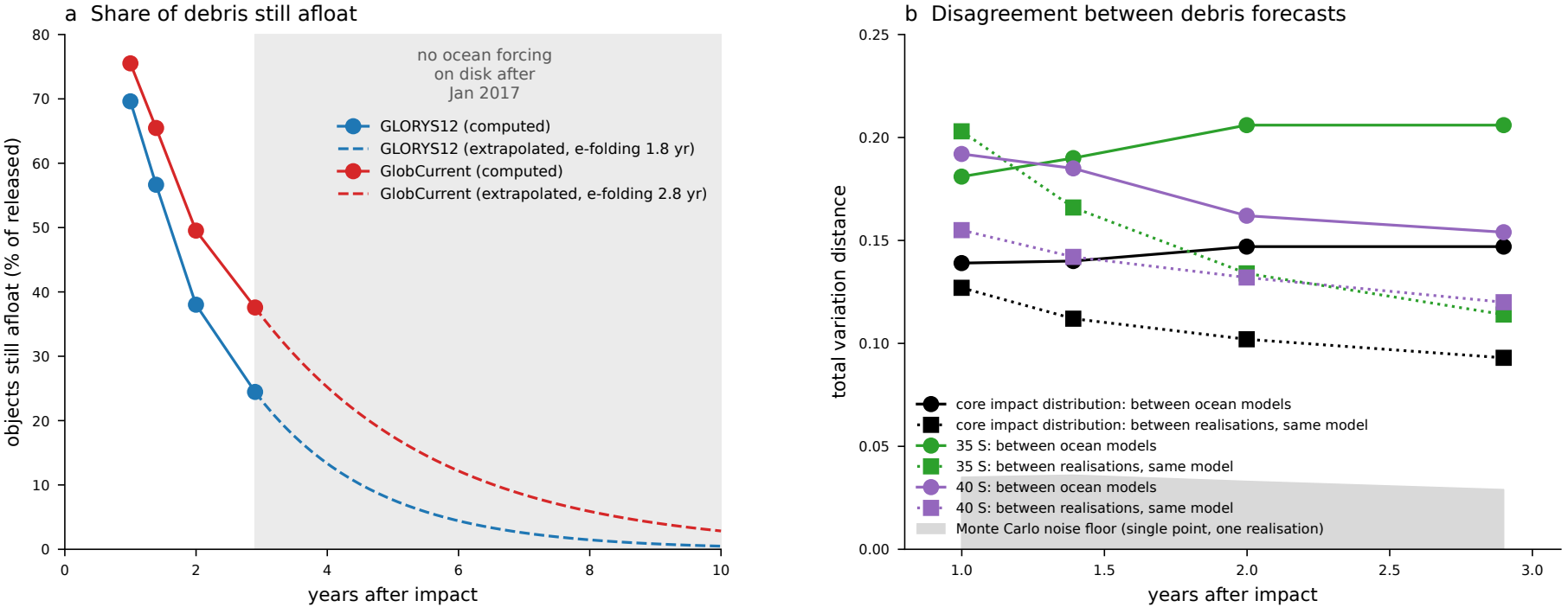
How far ahead the debris forecast can go, and how uncertain it becomes



Plain English. (a) The share of objects that are still afloat falls as objects reach coasts. At 1 year the two ocean models differ by 6 percentage points; after that the GLORYS12 model puts objects ashore faster, and at 2.9 years they differ by 13 points. After January 2017 we have no ocean data on disk, so the dashed lines continue the trend only; they are not a drift calculation. The model has no sinking from marine growth or decay, so real debris afloat after 3 years will be fewer than shown. (b) The difference between two forecasts, as the share of objects that are in a different place (0 = the same, 1 = no overlap). Place means a 5 x 5 deg ocean box, a coast region, or outside the model area. The difference between the two ocean models stays at about 15-20 % and does not grow. The difference between realisations of the same model gets smaller with time. About 81 % of the debris that is ashore at 2.9 years was already ashore at 2 years; Australian arrivals are almost complete by 2 years, while African and Madagascar arrivals continue.

Technical. Impact PDF: run J waypoints end-of-flight posterior, 00:19 option D0', 'D0 total q=0.333 (C2 silent)', seeds 1-8 of 24 evidence-weighted, 200 draws, ~10,000 particles each per model; 35 S / 40 S: single 7th-arc points, 100,008 particles per model. Drift: stand-in `snapshots` harness on d51d80a, 15-160 E, refloat off, 6 h step, K log-uniform 100-1000 m2/s, ocean error per environment seed 1-4; GLORYS12V1 0.494 m + ERA5; GlobCurrent 0 m + ERA5, windage U(-0.75,-0.60) % U10. (a) Exponential fit of afloat share at 1.4-2.9 yr; extrapolation ignores domain exit (10-18 % by 2.9 yr for the core PDF) and is not a forecast. (b) TV on the state vector {afloat by 5 deg box, beached by coast region, other}; realisation TV = mean over the 6 within-model pairs of environment seeds, both models; floor = mean TV of two multinomial resamples at one-seed size of a single release.